AGGREGATING LOGNORMAL RISKS IN GROUPS: JOINT TAILS AND COPULAS

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Abstract. In this contribution we investigate simultaneous large values of sums and maxima of dependent lognormal risks partitioned into finitely many groups. We obtain sharp joint-tail asymptotics and show that the two tails are asymptotically proportional, extending the scalar single-big-jump principle. It turns out that the copulas of the group sums and maxima are asymptotically independent and have the same logarithmic upper-tail exponents. The joint extreme-value copula of the sums and maxima has complete dependence within each group and independence between groups. For two groups, we give a sharp copula-tail equivalent when Gaussian margins are identical within each group, and establish hidden regular variation for common Gaussian variance and arbitrary means. We also obtain conditional aggregation and common-excess limits, and extend tail proportionality to signed sums under nondegeneracy assumptions. A log-elliptical extension distinguishes finite and diverging conditional residual scales.

1. Introduction

Lognormal models are widely used to model positive risks subject to multiplicative variation. In insurance and finance, sums of dependent lognormal random variables arise in the modelling of aggregate losses and portfolio values, while weighted sums enter the pricing of arithmetic basket options [1–3]. The aggregation problem also has a long history in communication systems [4]. In wireless communications, sums of lognormal variables describe aggregate interference under lognormal shadowing [5, 6]. Lognormal models also occur in environmental science for pollutant concentrations [7] and in cellular biology for protein abundances across populations of bacterial cells [8]. These applications motivate the study of both individual lognormal quantities and their aggregates.

A basic result for a single aggregate is the single-big-jump principle. Specifically, for a Gaussian vector $(X_1, \dots, X_n)$ with positive definite covariance matrix, Asmussen and Rojas-Nandayapa [1] show that

$$\mathbb{P}\left\{\sum_{j=1}^{n} e^{X_j} > u\right\} \sim \mathbb{P}\left\{\max_{1\le j\le n} e^{X_j} > u\right\}, \qquad u \to \infty.$$

In our notation $f(u) \sim g(u)$ for two positive functions f, g means $\lim_{u\to\infty} f(u)/g(u) = 1$.

The above approximation shows that a large value of the sum is asymptotically accounted for by its largest summand. Related aggregation results have been established for log-elliptical risks [9], suitable classes of log-linear risks [10], and weighted sums of powers of Dirichlet risks [11]. For a general approach to the aggregation of rapidly varying risks, see [12].

The scalar principle also underlies efficient simulation of rare events for correlated lognormal sums [13]. For sharp tail asymptotics of linear combinations, including differences of correlated lognormal variables, see [3].

In many applications, risks are reported or managed in groups. Examples include losses in different business lines, values of several portfolios, and interference totals associated with different groups of transmitters. This leads to a vector of group sums and to questions about simultaneous large values of its components. The scalar single-big-jump principle applies to

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each group separately. To assess simultaneous exceedances, one needs the joint tail of the group totals and its relation to the joint tail of the corresponding group maxima.
We now introduce the setting. Let $A_1, \ldots, A_d$ be a partition of $\{1, \ldots, N\}$ into nonempty sets, where N and d are fixed positive integers. Suppose that $\boldsymbol{X}$ is an $\mathbb{R}^N$-valued Gaussian random vector with mean vector $\boldsymbol{\mu}$ and positive definite covariance matrix Σ, i.e.,

$$\boldsymbol{X} \sim \mathcal{N}_N(\boldsymbol{\mu}, \Sigma), \qquad \Sigma \succ 0 \tag{1.1}$$

and define the group sums and group maxima by

$$\boldsymbol{S} = \left(\sum_{j \in A_\ell} e^{X_j} \right)_{1 \le \ell \le d}, \qquad \boldsymbol{Q} = \left(\max_{j \in A_\ell} e^{X_j} \right)_{1 \le \ell \le d}. \tag{1.2}$$

Their continuous margins determine unique copulas, denoted by $C_{\boldsymbol{S}}$ and $C_{\boldsymbol{Q}}$, respectively. Throughout the paper, vector inequalities and elementary functions are interpreted componentwise, and $\mathbf{1}$ denotes the vector of ones of the required dimension. Unless explicitly stated otherwise, group indices ℓ, r belong to $\{1, \ldots, d\}$ and coordinate indices j, k belong to $\{1, \ldots, N\}$. Subvectors use the indices in their specified set. We write $\overline{\Phi}(x) = \mathbb{P}\{Z > x\}$ for $Z \sim \mathcal{N}(0,1)$, $\mathrm{Exp}(\lambda)$ for the exponential law with rate $\lambda > 0$, and $\boldsymbol{e}_j$ for the jth coordinate vector, $1 \le j \le N$. Our first question is the following.

Q1. *How does the single-big-jump principle change when several group sums must be large simultaneously?*

We show that under (1.1) there exists a finite constant $C_* \ge 1$ such that

$$\mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\} \sim C_* \mathbb{P}\{\boldsymbol{Q} > u\mathbf{1}\}, \qquad u \to \infty. \tag{1.3}$$

Theorem 2.1 gives sharp asymptotics for both probabilities and it identifies the constant C_*, which can be larger than one when $d > 1$, as illustrated in Examples 3.1 and 3.2. Its value is determined by the dominant Gaussian configurations and the conditional contributions of the remaining risks in groups on the boundary of the exceedance region. The constant also describes the probability that all group maxima exceed u, conditionally on simultaneous exceedance by the group sums. We obtain an exponential limit and mean asymptotics for the common excess above u. An extension to signed sums is established under the nondegeneracy conditions (3.14) in Proposition 3.6.
The dependence between the group totals raises a second question.

Q2. *What is the upper-tail dependence between group sums and how does it compare with that between group maxima?*

The distinction between asymptotic independence and residual dependence is central to this question. The dependence measures in [14, 15] quantify joint extremes beyond the usual upper tail-dependence coefficient. We show that $C_{\boldsymbol{S}}$ and $C_{\boldsymbol{Q}}$ are asymptotically independent and have the same joint upper-tail exponents. These exponents are determined by Gaussian quadratic optimisation problems. For each pair of groups, they yield an explicit residual dependence coefficient on the logarithmic scale. If the log-variance is constant within each of the two groups, this coefficient is $(1 + \rho_*)/2$, where ρ_* is the largest correlation between Gaussian coordinates belonging to different groups. This extends the familiar Gaussian residual dependence formula [16, 17] to group sums and maxima. For two groups with a common Gaussian variance across both groups, we also establish hidden regular variation after Pareto standardization, allowing arbitrary Gaussian means. When the upper tails of the group maxima are asymptotically equivalent across groups, as in Remark 2.6, C_* is also the limiting ratio of the joint upper-tail probabilities of the two copulas at common marginal probability levels. We also identify the extreme-value limit of the joint copula of $(\boldsymbol{S}, \boldsymbol{Q})$: each sum and its own maximum are completely dependent in the limit, while different groups are independent. This limit does not determine the finer asymptotics of simultaneous group exceedances.

The paper is organised as follows. Section 2 presents the sharp joint tail asymptotics and studies the copulas of group sums and maxima. Section 3 collects extensions and examples, including common excess, signed sums, conditional risk measures, and a log-elliptical extension with Gumbel radial tails. Section 4 collects the auxiliary results and proofs. The Gaussian orthant tail and conditional limit arguments build on [18, 19].

2. Main Results

We consider the Gaussian setting (1.1) and the group sums and maxima defined in (1.2). Put $I_N = \{1, \dots, N\}$. For a nonempty index set $I \subset I_N$, write $\boldsymbol{x}_I$ for the subvector with indices in I, Σ_{II} for the corresponding principal submatrix, and $\Sigma_{I_N,I}$ for the $N \times |I|$ matrix formed by the columns of Σ indexed by I. Indices are always listed in increasing order. Positive definiteness of Σ ensures that every Σ_{II} is invertible.

Consider nonempty index sets $I \subset I_N$ containing at most one coordinate from each group, so that $|I \cap A_\ell| \le 1$ for every $1 \le \ell \le d$. Define

$$\boldsymbol{\lambda}_I = \Sigma_{II}^{-1}\mathbf{1}_I, \qquad \boldsymbol{s}^I = \Sigma_{I_N,I}\boldsymbol{\lambda}_I, \qquad \kappa_I = \tfrac{1}{2}\mathbf{1}_I^{\mathsf{T}}\Sigma_{II}^{-1}\mathbf{1}_I. \tag{2.1}$$

Here $\boldsymbol{s}_I^I = \mathbf{1}_I$, and κ_I is the Gaussian quadratic cost of the configuration $\boldsymbol{s}^I$. Among these index sets we write

$$\begin{aligned} &\mathfrak{I} = \{I : \boldsymbol{\lambda}_I > \mathbf{0}_I, \quad \max_{j \in A_\ell} s_j^I \ge 1 \text{ for every } 1 \le \ell \le d\}, \\ &\kappa = \min_{I \in \mathfrak{I}} \kappa_I, \qquad \mathfrak{I}_* = \{I \in \mathfrak{I} : \kappa_I = \kappa\}. \end{aligned} \tag{2.2}$$

The set $\mathfrak{I}$ is nonempty by Lemma 4.1. The same lemma identifies $\boldsymbol{s}^I$, $I \in \mathfrak{I}_*$ as the configurations of minimum Gaussian quadratic cost subject to $\max_{j \in A_\ell} x_j \ge 1$ for every $1 \le \ell \le d$. For $I \in \mathfrak{I}_*$, define

$$\begin{aligned} B_I &= \{1 \le \ell \le d : I \cap A_\ell \ne \varnothing\}, \\ \mathcal{C}_I &= \{\ell \in \{1, \dots, d\} \setminus B_I : \max_{j \in A_\ell} s_j^I = 1\}, \\ J_{\ell,I} &= \{j \in A_\ell : s_j^I = 1\}, \qquad \ell \in \mathcal{C}_I. \end{aligned} \tag{2.3}$$

Thus B_I contains the groups represented by a selected coordinate, while $\mathcal{C}_I$ contains the unrepresented groups whose largest coordinate in $\boldsymbol{s}^I$ equals one. In each such boundary group, $J_{\ell,I}$ is nonempty and collects the coordinates that contribute at scale u under the conditional limit below. The remaining unrepresented groups have $\max_{j \in A_\ell} s_j^I > 1$.

For any nonempty $I \subset \{1, \dots, N\}$, write $I^c = \{1, \dots, N\} \setminus I$. When $I^c \ne \varnothing$, Gaussian regression uses

$$\begin{aligned} D_I &= \Sigma_{I^cI}\Sigma_{II}^{-1}, \qquad \Gamma_I = \Sigma_{I^cI^c} - D_I\Sigma_{II^c}, \\ \boldsymbol{d}_I &= \boldsymbol{\mu}_{I^c} - D_I\boldsymbol{\mu}_I, \quad \boldsymbol{Z}^{(I)} \sim \mathcal{N}(\mathbf{0}, \Gamma_I). \end{aligned} \tag{2.4}$$

The Schur complement Γ_I is positive definite. Thus $\boldsymbol{d}_I + \boldsymbol{Z}^{(I)}$ has the conditional law of $\boldsymbol{X}_{I^c}$ given $\boldsymbol{X}_I = \mathbf{0}_I$, as used below.

The contribution of the boundary groups is described by the random vectors $\boldsymbol{S}^{(I)}$ and $\boldsymbol{Q}^{(I)}$, indexed by $\mathcal{C}_I$, with joint law

$$\operatorname{Law}(\boldsymbol{S}^{(I)}, \boldsymbol{Q}^{(I)}) = \operatorname{Law}\left(\left(\sum_{j \in J_{\ell,I}} e^{X_j} \right)_{\ell \in \mathcal{C}_I}, \left(\max_{j \in J_{\ell,I}} e^{X_j} \right)_{\ell \in \mathcal{C}_I} \middle| \boldsymbol{X}_I = \mathbf{0}_I \right). \tag{2.5}$$

Here conditioning on $\boldsymbol{X}_I = \mathbf{0}_I$ means evaluating the Gaussian conditional distribution at zero. For every $I \in \mathfrak{I}_*$, we have the following weak convergence (see the proof of Theorem 2.1)

$$\operatorname{Law}\left(\frac{\boldsymbol{S}_{\mathcal{C}_I}}{u}, \frac{\boldsymbol{Q}_{\mathcal{C}_I}}{u} \middle| \boldsymbol{X}_I > (\log u)\mathbf{1}_I \right) \xrightarrow[u \to \infty]{\mathrm{w}} \operatorname{Law}(\boldsymbol{S}^{(I)}, \boldsymbol{Q}^{(I)}). \tag{2.6}$$

The limiting margins have no atom at one, so the corresponding conditional exceedance probabilities converge to $\mathbb{P}\Big\{\boldsymbol{S}^{(I)} > \mathbf{1}\Big\}$ and $\mathbb{P}\Big\{\boldsymbol{Q}^{(I)} > \mathbf{1}\Big\}$. If $\mathcal{C}_I = \varnothing$, both of these probabilities are defined to equal one.
Throughout, $\boldsymbol{T} = (T_\ell)_{1\le \ell\le d} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$, and $\boldsymbol{T}^{(I)}$ is the corresponding boundary vector in (2.5). Let $\mathfrak{H}$ consist of all $I \in \mathfrak{I}_*$ maximizing $\boldsymbol{\lambda}_I^\mathsf{T}\boldsymbol{\mu}_I$ and, among these, minimizing $|I|$, and put

$$A_I = \frac{\exp(-\boldsymbol{\mu}_I^\mathsf{T}\Sigma_{II}^{-1}\boldsymbol{\mu}_I/2)}{(2\pi)^{|I|/2}\sqrt{\det \Sigma_{II}}\prod_{j\in I}\lambda_{I,j}}, \qquad I \in \mathfrak{I}_*.$$

The following theorem combines the dominant contributions.

Theorem 2.1. *As $u \to \infty$ we have*

$$\mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\} \sim \sum_{I\in\mathfrak{I}_*} p_I(u)\mathbb{P}\{\boldsymbol{T}^{(I)} > \mathbf{1}\}, \tag{2.7}$$

where $p_I(u) = \mathbb{P}\{\boldsymbol{X}_I > (\log u)\mathbf{1}_I\}$, $I \in \mathfrak{I}_*$.

Remark 2.2. *The constant in (1.3) is given by*

$$C_* = \frac{\sum_{I\in\mathfrak{H}} A_I\mathbb{P}\{\boldsymbol{S}^{(I)} > \mathbf{1}\}}{\sum_{I\in\mathfrak{H}} A_I\mathbb{P}\{\boldsymbol{Q}^{(I)} > \mathbf{1}\}} \in [1,\infty). \tag{2.8}$$

If $d = 1$, then $C_* = 1$. For $d = 2$ with common Gaussian variance, a singleton $I = \{j\}$ has $s_k^I = \Sigma_{kj}/\Sigma_{jj} < 1$ for every $k \ne j$, by positive definiteness. It therefore cannot meet the constraint in the other group. Thus every $I \in \mathfrak{I}_*$ represents both groups and $\mathcal{C}_I = \varnothing$, giving $C_* = 1$ also in this case. For $d \ge 3$, equal variances can still give $C_* > 1$; see Example 3.2.

Next, for $1 \le \ell \le d$, let $F_{\ell,\boldsymbol{T}}$ be the cdf of T_ℓ. The positive Gaussian density ensures that these margins are continuous and strictly increasing on $(0,\infty)$. The joint law therefore has the unique copula

$$C_{\boldsymbol{T}}(\boldsymbol{v}) = \mathbb{P}\Big\{T_\ell \le F_{\ell,\boldsymbol{T}}^{-1}(v_\ell),\ 1 \le \ell \le d\Big\}, \qquad \boldsymbol{v} \in (0,1)^d. \tag{2.9}$$

Write $\sigma_j = \sqrt{\Sigma_{jj}}$, $1 \le j \le N$, and $\rho_{jk} = \Sigma_{jk}/(\sigma_j\sigma_k)$, $1 \le j,k \le N$. In order to compare joint upper tails at a common marginal probability level $q \in (0,1)$, put

$$J_{B,\boldsymbol{T}}(q) = \mathbb{P}\Big\{T_\ell > F_{\ell,\boldsymbol{T}}^{-1}(1-q),\ \ell \in B\Big\}, \qquad \varnothing \ne B \subset \{1,\dots,d\}. \tag{2.10}$$

Proposition 2.3. *Put $\sigma_\ell^* = \max_{j\in A_\ell}\sigma_j$, $1 \le \ell \le d$. For every nonempty $B \subset \{1,\dots,d\}$*

$$J_{B,\boldsymbol{T}}(q) = q^{\beta_B + o(1)}, \qquad q \downarrow 0, \tag{2.11}$$

where

$$\beta_B = \min_{\boldsymbol{x}\in\mathbb{R}^N}\left\{\boldsymbol{x}^\mathsf{T}\Sigma^{-1}\boldsymbol{x} : \max_{j\in A_\ell} x_j \ge \sigma_\ell^*,\ \ell \in B\right\}. \tag{2.12}$$

We have $\beta_{\{\ell\}} = 1$, $1 \le \ell \le d$, and $\beta_B > 1$ for $|B| \ge 2$, so both copulas are asymptotically independent. For distinct $\ell, r \in \{1,\dots,d\}$ with constant Gaussian variance within each group,

$$\beta_{\{\ell,r\}} = \frac{2}{1 + \max_{a\in A_\ell,\ b\in A_r}\rho_{ab}}. \tag{2.13}$$

For distinct ℓ, r, (2.11) gives the residual dependence coefficient on the logarithmic scale:

$$\eta_{\ell r,\boldsymbol{T}} := \lim_{q\downarrow 0}\frac{\log q}{\log J_{\{\ell,r\},\boldsymbol{T}}(q)} = \frac{1}{\beta_{\{\ell,r\}}}.$$

When the Gaussian variances are constant within each group, (2.13) gives $\eta_{\ell r,\boldsymbol{T}} = (1 + \max_{a\in A_\ell, b\in A_r}\rho_{ab})/2$. For singleton groups, this is the Gaussian residual dependence formula in [16] and [17, Section 2]. For residual and weak tail dependence, see, e.g., [14, 15, 20]. The next result gives a sharp equivalent when the Gaussian margins are identical within each of the two groups.

Corollary 2.4. *Let $d \geq 2$. For $i = 1, 2$, suppose $\mu_j = m_i$ and $\Sigma_{jj} = s_i^2$ for $j \in A_i$, where $m_i \in \mathbb{R}$ and $s_i > 0$. Put $n_i = |A_i|$, $i = 1, 2$, and*

$$\rho_* = \max_{a \in A_1,\ b \in A_2} \rho_{ab}, \qquad n_* = \#\{(a, b) \in A_1 \times A_2 : \rho_{ab} = \rho_*\}.$$

For $B = \{1, 2\}$, as $q \downarrow 0$

$$J_{B,\boldsymbol{T}}(q) \sim K_* q^{2/(1+\rho_*)} [\log(1/q)]^{-\rho_*/(1+\rho_*)}, \tag{2.14}$$

where

$$K_* = \frac{n_*(1+\rho_*)^2}{\sqrt{1-\rho_*^2}} (n_1 n_2)^{-1/(1+\rho_*)} (4\pi)^{-\rho_*/(1+\rho_*)}. \tag{2.15}$$

For the next result take $d = 2$ and put $W_{i,\boldsymbol{T}} = (1 - F_{i,\boldsymbol{T}}(T_i))^{-1}$, $i = 1, 2$. For hidden regular variation under Gaussian copulas, see [21, 22].

Corollary 2.5. *If the Gaussian coordinates have common variance $s^2 > 0$ and arbitrary means, then $(W_{1,\boldsymbol{T}}, W_{2,\boldsymbol{T}})$ is regularly varying with index one on $[0, \infty)^2 \setminus \{(0, 0)\}$ and has hidden regular variation on $(0, \infty)^2$ with index $\alpha_0 = 2/(1 + \rho_*)$, where $\rho_* = \max_{a \in A_1,\ b \in A_2} \rho_{ab}$.*

Remark 2.6. *If $\mathbb{P}\{Q_\ell > u\} \sim \mathbb{P}\{Q_1 > u\}$ as $u \to \infty$ for $1 \leq \ell \leq d$, then*

$$\frac{J_{B,\boldsymbol{S}}(q)}{J_{B,\boldsymbol{Q}}(q)} \longrightarrow C_*, \qquad q \downarrow 0, \quad B = \{1, \ldots, d\}. \tag{2.16}$$

Let $C_{\boldsymbol{S},\boldsymbol{Q}}$ denote the copula of $(S_1, \ldots, S_d, Q_1, \ldots, Q_d)$.

Corollary 2.7. *Under (1.1), for $\boldsymbol{x}, \boldsymbol{y} \in [0, \infty)^d$,*

$$\lim_{q \downarrow 0} \frac{1 - C_{\boldsymbol{S},\boldsymbol{Q}}(\mathbf{1} - q\boldsymbol{x}, \mathbf{1} - q\boldsymbol{y})}{q} = \sum_{\ell=1}^{d} \max(x_\ell, y_\ell). \tag{2.17}$$

Consequently, $C_{\boldsymbol{S},\boldsymbol{Q}}$ is in the max-domain of attraction of the extreme-value copula

$$C_{\mathrm{EV}}(\boldsymbol{u}, \boldsymbol{v}) = \prod_{\ell=1}^{d} \min(u_\ell, v_\ell), \qquad \boldsymbol{u}, \boldsymbol{v} \in [0, 1]^d. \tag{2.18}$$

Thus the limiting sum–maximum pairs are completely dependent within each group and independent between groups.

The limit (2.17) describes exceedance unions at marginal probability scale q. Simultaneous exceedances of two or more groups have probability $o(q)$ by Proposition 2.3, so their relative asymptotics are not determined by (2.18). In particular, the conditional probability in (3.1) can still converge to $C_*^{-1} < 1$, as in Examples 3.1 and 3.2. The latter example has equal variances and $d = 3$. Complete upper-tail dependence within each individual pair therefore coexists with a nontrivial sum–maximum ratio under simultaneous conditioning.

3. Examples and Extensions

The constant C_* is strictly larger than one precisely when there exist $I \in \mathfrak{H}$ and $\ell \in \mathcal{C}_I$ with $|J_{\ell,I}| \geq 2$. Indeed, full support of the conditional Gaussian law then gives positive probability that this boundary sum exceeds one while its maximum is below one and all other boundary maxima exceed one; if every $J_{\ell,I}$ is a singleton, the boundary sums and maxima agree. Formula (2.8) proves the criterion, since all weights are positive. It also has the conditional interpretation

$$\mathbb{P}\{\boldsymbol{Q} > u\mathbf{1} \mid \boldsymbol{S} > u\mathbf{1}\} \longrightarrow C_*^{-1}, \qquad u \to \infty. \tag{3.1}$$

Example 3.1. *For independent standard normals Y_0, Y_1, Y_2, take $\boldsymbol{X} = (Y_0, Y_0+Y_1, Y_0+Y_2)^\mathsf{T}$, $A_1 = \{1\}$ and $A_2 = \{2,3\}$. Thus*

$$\boldsymbol{S} = (e^{Y_0}, e^{Y_0+Y_1} + e^{Y_0+Y_2})^\mathsf{T} \tag{3.2}$$

and hence $C_* = \frac{4}{3}\mathbb{P}\Big\{e^{Y_1} + e^{Y_2} > 1\Big\} \in (1, 4/3)$, with numerical value $C_* \approx 1.182066$.

Example 3.2. *Let G_0, G_1, G_2 and $(\varepsilon_{i,k})_{1\le i\le 2, k\in\mathbb{Z}}$ be independent standard normals and set*

$$X_i(k) = \frac{G_0}{2} + \frac{G_1\cos(k\pi/3) + G_2\sin(k\pi/3)}{\sqrt{2}} + \frac{\varepsilon_{i,k}}{2}, \qquad i \in \{1,2\}, \quad k \in \mathbb{Z}. \tag{3.3}$$

This stationary Gaussian process has unit variances and positive definite covariances for finite samples of distinct coordinate–time pairs. For the three groups $S_k = e^{X_1(k)} + e^{X_2(k)}$, $k = 0, 1, 2$, we have $1 < C_* < 3/2$, with numerical value $C_* \approx 1.332802$. Their identical marginal laws also give the copula ratio (2.16). For every pair of these groups, $C_* = 1$ and the two aggregate copulas have the same sharp joint-tail equivalent in (2.14).

Remark 3.3. *Let $(X_1(k), X_2(k))$, $k \in \mathbb{Z}$, be stationary Gaussian with means μ_i, $i = 1, 2$, and common variance $v > 0$. On finite nonempty grids $T_i \subset \mathbb{Z}$, put $S_i = \sum_{k\in T_i} e^{X_i(k)}$ and $Q_i = \max_{k\in T_i} e^{X_i(k)}$, $i = 1, 2$. Assume a positive definite sampled covariance matrix, and put*

$$c_* = \max_{(k,l)\in T_1\times T_2} \mathrm{Cov}(X_1(k), X_2(l)) \tag{3.4}$$

and let n_* count the pairs attaining this maximum. For a Gaussian pair $\boldsymbol{Y}$ with mean $(\mu_1, \mu_2)^\mathsf{T}$ and covariance $\left(\begin{smallmatrix} v & c_* \\ c_* & v \end{smallmatrix}\right)$,

$$\mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\} \sim \mathbb{P}\{\boldsymbol{Q} > u\mathbf{1}\} \sim n_*\mathbb{P}\{\boldsymbol{Y} > (\log u)\mathbf{1}\}, \qquad u \to \infty. \tag{3.5}$$

The copula equivalent in Corollary 2.4 holds with $n_i = |T_i|$, $i \in \{1,2\}$, $\rho_* = c_*/v$ and this n_*.

More generally, a finite Gaussian sample may be grouped by component or by observation time. Theorem 2.1 applies whenever its covariance is positive definite; stationarity is not required.

Example 3.4 (Equicorrelated Gaussian risks)**.** *Let $N \ge 2$, $\boldsymbol{\mu} = m\mathbf{1}$ and*

$$\Sigma = \sigma^2[(1-\rho)\mathrm{Id}_N + \rho\mathbf{1}\mathbf{1}^\mathsf{T}], \qquad \sigma > 0, \quad -\frac{1}{N-1} < \rho < 1,$$

where Id_N is the identity matrix. Put $n_\ell = |A_\ell|$ and $D_k = 1 + (k-1)\rho$, $1 \le k \le N$. Every admissible selector contains one coordinate from each group, and all $\prod_{\ell=1}^d n_\ell$ such selectors are dominant. Hence

$$\kappa = \frac{d}{2\sigma^2 D_d}, \qquad C_* = 1.$$

Writing $z = (\log u - m)/\sigma$, as $u \to \infty$,

$$\mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\} \sim \mathbb{P}\{\boldsymbol{Q} > u\mathbf{1}\} \sim \left(\prod_{\ell=1}^d n_\ell\right) \frac{D_d^{d-1/2}}{(2\pi)^{d/2}(1-\rho)^{(d-1)/2}} z^{-d}\exp\left(-\frac{dz^2}{2D_d}\right). \tag{3.6}$$

For any nonempty $B \subset \{1, \dots, d\}$ with $b = |B|$,

$$\beta_B = \frac{b}{1 + (b-1)\rho}. \tag{3.7}$$

Thus the joint extreme-value copula (2.18) is the same for every admissible ρ, while the simultaneous upper-tail exponents vary with ρ.

3.1. **Aggregation and common excess given exceedance.** We return to the positive group sums and maxima in (1.2).

Remark 3.5. *Put* $S_\wedge = \min_{1\le\ell\le d} S_\ell$ *and* $R_u = S_\wedge / \min_{1\le\ell\le d} Q_\ell$. *Conditionally on* $\boldsymbol{S} > u\mathbf{1}$,

$$R_u \xrightarrow[u\to\infty]{\mathrm{w}} \mathcal{R}, \qquad \mathbb{P}\{\mathcal{R}=1\} = C_*^{-1}. \tag{3.8}$$

Thus $C_* = 1$ *iff* $R_u \to 1$ *in conditional probability. Moreover, we have*

$$\mathbb{P}\{\mathcal{R}\le r\} = \frac{\sum_{I\in\mathfrak{H}} A_I \mathbb{P}\Big\{\boldsymbol{S}^{(I)} > \mathbf{1},\ \boldsymbol{Q}^{(I)} > r^{-1}\mathbf{1}\Big\}}{\sum_{I\in\mathfrak{H}} A_I \mathbb{P}\Big\{\boldsymbol{S}^{(I)} > \mathbf{1}\Big\}}, \qquad r\ge 1, \tag{3.9}$$

with $\mathbb{P}\{\mathcal{R}\le r\} = 0$ *for* $r < 1$.
Furthermore, as $u\to\infty$,

$$\mathrm{Law}\left(\frac{2\kappa\log u}{u}(S_\wedge - u)\,\middle|\,\boldsymbol{S} > u\mathbf{1}\right) \xrightarrow{\mathrm{w}} \mathrm{Exp}(1), \tag{3.10}$$

$$\mathbb{E}\{S_\wedge - u \mid \boldsymbol{S} > u\mathbf{1}\} \sim \frac{u}{2\kappa\log u}. \tag{3.11}$$

For applications to basket options and options on minima, see [2, 23].

3.2. **Signed sums.** For $1\le\ell\le d$, split $A_\ell = P_\ell \dot\cup N_\ell$, with $P_\ell \ne \varnothing$ and define

$$\widetilde{S}_\ell = \sum_{j\in P_\ell} e^{X_j} - \sum_{k\in N_\ell} e^{X_k}, \qquad \widetilde{Q}_\ell \ = \max_{j\in P_\ell} e^{X_j} - \sum_{k\in N_\ell} e^{X_k}. \tag{3.12}$$

Write $\widetilde{\boldsymbol{S}} = (\widetilde{S}_\ell)_{1\le\ell\le d}$ and $\widetilde{\boldsymbol{Q}} = (\widetilde{Q}_\ell)_{1\le\ell\le d}$. Nonzero deterministic weights can be absorbed into the means and signs. With $\max\varnothing = -\infty$, put

$$\mathcal{F}_\pm = \left\{\boldsymbol{x}\in\mathbb{R}^N : \max_{j\in P_\ell} x_j \ge \max\left(1, \max_{k\in N_\ell} x_k\right),\ 1\le\ell\le d\right\}, \qquad \mathcal{S}_\pm \ = \operatorname*{argmin}_{\boldsymbol{x}\in\mathcal{F}_\pm} \tfrac{1}{2}\boldsymbol{x}^{\mathsf{T}}\Sigma^{-1}\boldsymbol{x}. \tag{3.13}$$

For each $\boldsymbol{s}\in\mathcal{S}_\pm$, write $\boldsymbol{g} = \Sigma^{-1}\boldsymbol{s}$ and $h_\ell = \max_{j\in P_\ell} s_j$, $1\le\ell\le d$. The nondegeneracy assumption is that, for every $1\le\ell\le d$ with $\boldsymbol{g}_{A_\ell} \ne \mathbf{0}$,

$$h_\ell = 1 \implies \sum_{j\in A_\ell} g_j > 0, \qquad s_k = h_\ell \implies g_k < 0 \quad (k\in N_\ell). \tag{3.14}$$

Proposition 3.6. *Under* (3.14), *for some* $C_\pm \in [1,\infty)$

$$\mathbb{P}\Big\{\widetilde{\boldsymbol{S}} > u\mathbf{1}\Big\} \sim C_\pm \mathbb{P}\Big\{\widetilde{\boldsymbol{Q}} > u\mathbf{1}\Big\}, \qquad u\to\infty. \tag{3.15}$$

For $d = 1$, $C_\pm = 1$.

Without negative terms this reduces to (1.3). For the scalar case, see also [3, Theorem 2] under its nondegeneracy assumptions. Under Proposition 3.6, the common-excess conclusions (3.10)–(3.11) also hold with $S_\wedge$ replaced by $\min_{1\le\ell\le d}\widetilde{S}_\ell$, conditioning on $\widetilde{\boldsymbol{S}} > u\mathbf{1}$, and with $\kappa_\pm = \min_{\boldsymbol{x}\in\mathcal{F}_\pm} \frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\Sigma^{-1}\boldsymbol{x}$ in place of κ.

3.3. **Conditional risk measures.** Use the Gaussian sums in (1.1)–(1.2); for related conditional risk measures, see [24]. For $\varnothing \ne B \subset \{1,\dots,d\}$ and $u > 0$, put $\mathcal{E}_B(u) = \{S_r > u,\ r\in B\}$. Write $\mathbb{P}_{\boldsymbol{\nu}}$ for probability under $\mathcal{N}_N(\boldsymbol{\nu},\Sigma)$, $\boldsymbol{\nu}\in\mathbb{R}^N$, and put $\boldsymbol{\mu}^{(j)} = \boldsymbol{\mu} + \Sigma\boldsymbol{e}_j$, $1\le j\le N$. Gaussian tilting gives, for $1\le\ell\le d$,

$$\mathbb{E}\{S_\ell \mid \mathcal{E}_B(u)\} = \sum_{j\in A_\ell} e^{\mu_j + \Sigma_{jj}/2} \frac{\mathbb{P}_{\boldsymbol{\mu}^{(j)}}\{\mathcal{E}_B(u)\}}{\mathbb{P}_{\boldsymbol{\mu}}\{\mathcal{E}_B(u)\}}. \tag{3.16}$$

Theorem 2.1 gives a sharp equivalent by evaluating the tails for the groups in B at each corresponding mean. For $B = \{1,\dots,d\}$ this gives the conditional tail expectation; for $\ell\notin B$, the marginal expected shortfall under stress in B.

Under Remark 3.3, put $n_i = |T_i|$, $i = 1, 2$. For $i, j \in \{1, 2\}$ with $i \neq j$ and $u > 0$, define the marginal expected shortfall and marginal mean excess by

$$\mathrm{MES}_i(u) = \mathbb{E}\{S_i \mid S_j > u\}, \qquad \mathrm{MME}_i(u) = \mathbb{E}\{(S_i - u)_+ \mid S_j > u\},$$

where $x_+ = \max(x, 0)$. For these measures under regular variation and asymptotic tail independence, see [25]. For truncated moments, including those of multivariate lognormal vectors, see [26].

Remark 3.7. *As $u \to \infty$ we have*

$$\mathrm{MES}_i(u) \sim \frac{n_*}{n_j} \exp\left(\mu_i + \frac{v}{2} - \frac{c_* \mu_j}{v} - \frac{c_*^2}{2v}\right) u^{c_*/v}, \tag{3.17}$$

$$\mathrm{MME}_i(u) \sim (v + c_*) \frac{u}{\log u} \frac{\mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\}}{\mathbb{P}\{S_j > u\}}. \tag{3.18}$$

The tail ratio in (3.18) follows from (3.5) and $\mathbb{P}\{S_j > u\} \sim n_j \overline{\Phi}((\log u - \mu_j)/\sqrt{v})$.

3.4. **Log-elliptical sums.** Only in this subsection and its proof, replace (1.1) by

$$\boldsymbol{X} = \boldsymbol{\mu} + RA\boldsymbol{U}_N, \qquad \Sigma = AA^{\mathsf{T}} \succ 0, \qquad R \perp \boldsymbol{U}_N, \qquad \boldsymbol{U}_N \sim \mathrm{Unif}(\mathbb{S}^{N-1}), \tag{3.19}$$

where $A \in \mathbb{R}^{N \times N}$ and $\mathbb{S}^{N-1}$ is the unit sphere in $\mathbb{R}^N$. The matrix Σ is now a scatter matrix. Use the positive aggregates (1.2) and the selector notation in (2.1)–(2.3), and put $p_I(u) = \mathbb{P}\{\boldsymbol{X}_I > (\log u)\mathbf{1}_I\}$, $I \in \mathfrak{I}_*$, under (3.19). Suppose $R > 0$ has infinite upper endpoint and survival function $\overline{F}$ in the Gumbel max-domain of attraction: for a positive measurable scaling function w

$$\frac{\overline{F}(r + x/w(r))}{\overline{F}(r)} \longrightarrow e^{-x}, \qquad r \to \infty, \quad x \in \mathbb{R}. \tag{3.20}$$

Write $w \in \mathrm{RV}_a$ if $w(cr)/w(r) \to c^a$ for every $c > 0$. For $\tau > 0$, define $\boldsymbol{S}_\tau^{(I)}, \boldsymbol{Q}_\tau^{(I)}$ by (2.5) using the Gaussian reference law $\mathcal{N}_N(\boldsymbol{\mu}, \tau^2\Sigma)$. Also let $\boldsymbol{Z}^{(I)} \sim \mathcal{N}_{N-|I|}(\mathbf{0}, \Gamma_I)$ for $I \in \mathfrak{I}_*$ with $I^c \neq \varnothing$, where Γ_I is as in (2.4). Boundary probabilities are one when $\mathcal{C}_I = \varnothing$.

Proposition 3.8. (i) *If $r/w(r) \to \tau^2 \in (0, \infty)$, then*

$$\mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\} \sim \sum_{I \in \mathfrak{I}_*} p_I(u) \mathbb{P}\left\{\boldsymbol{T}_\tau^{(I)} > \mathbf{1}\right\}, \qquad u \to \infty. \tag{3.21}$$

Moreover, $\lim_{u\to\infty} \mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\}/\mathbb{P}\{\boldsymbol{Q} > u\mathbf{1}\} = C_\tau$, where $C_\tau \in [1, \infty)$ is the constant (2.8) for the Gaussian reference law $\mathcal{N}_N(\boldsymbol{\mu}, \tau^2\Sigma)$.
(ii) *If $w \in \mathrm{RV}_a$ with $-1 < a < 1$, then*

$$\mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\} \sim \mathbb{P}\{\boldsymbol{Q} > u\mathbf{1}\} \sim \sum_{I \in \mathfrak{I}_*} p_I(u) \mathbb{P}\left\{\max_{j \in J_{\ell,I}} Z_j^{(I)} > 0, \ \ell \in \mathcal{C}_I\right\}, \qquad u \to \infty. \tag{3.22}$$

The active and residual scales are $1/w(r)$ and $\sqrt{r/w(r)}$. The latter converges to τ in part (i) and diverges in part (ii). Gumbel attraction alone need not give a fixed boundary constant; the collapsing regime $r/w(r) \to 0$ requires separate boundary analysis. The radial tail remains in the sharp orthant equivalent (4.28); it is not replaced by a Gaussian tail.
A further direction is to consider log-Liouville models, with $\boldsymbol{X} = \boldsymbol{\mu} + R\boldsymbol{D}$, where $R > 0$ is independent of a Dirichlet vector $\boldsymbol{D}$ with positive parameters on the positive L^1 unit sphere $\{\boldsymbol{x} \in [0, \infty)^N : \sum_{j=1}^N x_j = 1\}$. This representation suggests a similar radial analysis, with the simplex geometry requiring separate treatment of the boundary terms. For extremal attractors of Liouville copulas and their survival counterparts, see [27].

4. Further Results and Proofs

4.1. **Auxiliary facts.** We next present some auxiliary results, writing $t = \log u$ and $E_u = \{\boldsymbol{S} > u\mathbf{1}\}$. For $I \in \mathfrak{I}_*$ set

$$\boldsymbol{z}_I(t) = t\mathbf{1}_I - \boldsymbol{\mu}_I, \qquad \boldsymbol{g}_I(t) = \Sigma_{II}^{-1}\boldsymbol{z}_I(t), \qquad r_I(t) = (\boldsymbol{z}_I(t)^{\mathsf{T}}\Sigma_{II}^{-1}\boldsymbol{z}_I(t))^{1/2}. \tag{4.1}$$

For any nonempty $I \subset I_N$ we write

$$\boldsymbol{\Delta}_I = \boldsymbol{X}_I - t\mathbf{1}_I, \qquad \boldsymbol{\eta}_I = \boldsymbol{X}_{I^c} - \boldsymbol{\mu}_{I^c} - D_I(\boldsymbol{X}_I - \boldsymbol{\mu}_I). \tag{4.2}$$

The residual $\boldsymbol{\eta}_I$ is independent of $\boldsymbol{X}_I$ with law $\mathcal{N}(\mathbf{0}, \Gamma_I)$. Empty residuals are omitted. Put

$$\Omega = \Sigma^{-1}, \qquad q(\boldsymbol{x}) = \tfrac{1}{2}\boldsymbol{x}^{\mathsf{T}}\Omega\boldsymbol{x}, \qquad \mathcal{F} = \{\boldsymbol{x} \in \mathbb{R}^N : \max_{j\in A_\ell} x_j \geq 1, 1 \leq \ell \leq d\}. \tag{4.3}$$

Lemma 4.1.

$$\mathfrak{I} \neq \varnothing, \qquad \min_{\boldsymbol{x}\in\mathcal{F}} q(\boldsymbol{x}) = \kappa, \qquad \operatorname*{argmin}_{\boldsymbol{x}\in\mathcal{F}} q(\boldsymbol{x}) = \{\boldsymbol{s}^I : I \in \mathfrak{I}_*\}.$$

Moreover, we have

$$s_k^I < 1, \qquad I \in \mathfrak{I}_*, \quad \ell \in B_I, \quad k \in A_\ell \setminus I. \tag{4.4}$$

For the following asymptotic equivalence and lemma, let $\varnothing \neq I \subset \{1, \dots, N\}$ satisfy $\boldsymbol{\lambda}_I = \Sigma_{II}^{-1}\mathbf{1}_I > \mathbf{0}_I$. For bounded $\boldsymbol{\beta}_I(t)$, put

$$\boldsymbol{z}_I(t) = t\mathbf{1}_I + \boldsymbol{\beta}_I(t), \qquad \boldsymbol{g}_I(t) = \Sigma_{II}^{-1}\boldsymbol{z}_I(t), \qquad r_I(t)^2 = \boldsymbol{z}_I(t)^{\mathsf{T}}\boldsymbol{g}_I(t).$$

The choice $\boldsymbol{\beta}_I(t) = -\boldsymbol{\mu}_I$ recovers (4.1). The Gaussian orthant equivalent [18] is

$$\mathbb{P}\{\boldsymbol{X}_I > \boldsymbol{\mu}_I + \boldsymbol{z}_I(t)\} \sim \frac{e^{-r_I(t)^2/2}}{(2\pi)^{|I|/2}\sqrt{\det\Sigma_{II}}\prod_{j\in I} g_{I,j}(t)}, \qquad u \to \infty. \tag{4.5}$$

Lemma 4.2. *With the preceding notation and the residuals in* (4.2)

$$\operatorname{Law}\left(t(\boldsymbol{X}_I - \boldsymbol{\mu}_I - \boldsymbol{z}_I(t)), \boldsymbol{\eta}_I \mid \boldsymbol{X}_I > \boldsymbol{\mu}_I + \boldsymbol{z}_I(t)\right) \xrightarrow[u\to\infty]{\mathrm{w}} \left(\bigotimes_{j\in I} \operatorname{Exp}(\lambda_{I,j})\right) \otimes \mathcal{N}_{N-|I|}(\mathbf{0}, \Gamma_I). \tag{4.6}$$

If $I^c = \varnothing$, then we omit the residuals and Gaussian factor.

Proof of Lemma 4.1. Use the notation in (4.3). The feasible set is the finite union of the orthants $\{\boldsymbol{x} : \boldsymbol{x}_K \geq \mathbf{1}_K\}$ obtained by choosing one coordinate from each group. On each orthant, strict convexity and coercivity give a unique minimizer $\boldsymbol{s}$. The optimality conditions are

$$(\Omega\boldsymbol{s})_j \geq 0 \ (j \in K), \qquad (\Omega\boldsymbol{s})_j = 0 \ (j \notin K \text{ or } s_j > 1).$$

Its positive support $I = \{j : (\Omega\boldsymbol{s})_j > 0\}$ is nonempty, since $\Omega\boldsymbol{s} = \mathbf{0}$ would give the infeasible point $\boldsymbol{s} = \mathbf{0}$. Moreover,

$$\boldsymbol{s}_I = \mathbf{1}_I, \qquad (\Omega\boldsymbol{s})_I = \Sigma_{II}^{-1}\mathbf{1}_I > \mathbf{0}_I, \qquad \boldsymbol{s} = \Sigma_{I_N,I}\Sigma_{II}^{-1}\mathbf{1}_I = \boldsymbol{s}^I.$$

Thus $I \in \mathfrak{I}$. Conversely, every $I \in \mathfrak{I}$ gives a feasible point $\boldsymbol{s}^I$ with cost κ_I. Minimizing over the finitely many orthants proves all assertions about $\mathfrak{I}$, κ and the minimizers.
Finally, if $I \in \mathfrak{I}_*$ and another coordinate $k \neq j$ in the group of $j \in I$ satisfied $s_k^I \geq 1$, then $\boldsymbol{s}^I - \varepsilon\boldsymbol{e}_j$ would remain feasible, whereas

$$q(\boldsymbol{s}^I - \varepsilon\boldsymbol{e}_j) = \kappa - \varepsilon\lambda_{I,j} + \tfrac{1}{2}\varepsilon^2\Omega_{jj} < \kappa$$

for small $\varepsilon > 0$. This proves (4.4). □

Proof of (4.5) *and Lemma 4.2.* Put $K=\Sigma_{II}$ and $\boldsymbol{g}=\boldsymbol{g}_I(t)$. Translating the Gaussian density to the threshold gives

$$\mathbb{P}\{\boldsymbol{X}_I>\boldsymbol{\mu}_I+\boldsymbol{z}_I(t)\}=\frac{e^{-r_I(t)^2/2}}{(2\pi)^{|I|/2}\sqrt{\det K}}\int_{(0,\infty)^{|I|}} e^{-\boldsymbol{g}^\top\boldsymbol{x}-\boldsymbol{x}^\top K^{-1}\boldsymbol{x}/2}d\boldsymbol{x}. \tag{4.7}$$

Since $g_j/t\to\lambda_{I,j}>0$ for $j\in I$, the substitution $y_j=g_jx_j$, $j\in I$, makes the integrand converge to $e^{-\sum_{j\in I}y_j}$ and bounds it by that integrable function. Dominated convergence proves (4.5). With $\boldsymbol{x}=\boldsymbol{v}/t$, the normalized conditional density converges to $\prod_{j\in I}\lambda_{I,j}e^{-\lambda_{I,j}v_j}$ on the positive orthant. Scheffé's lemma gives the exponential-vector limit. Gaussian regression makes $\boldsymbol{\eta}_I\sim\mathcal{N}(\boldsymbol{0},\Gamma_I)$ independent of $\boldsymbol{X}_I$, so the convergence is joint with the unchanged residual law, proving (4.6). Empty residuals are omitted. □

4.2. **Proofs of the main result.**

Proof of Theorem 2.1. For $I\in\mathfrak{I}_*$, let $H_I(u)=\{\boldsymbol{X}_I>t\boldsymbol{1}_I\}$. By Lemma 4.2, conditionally on $H_I(u)$,

$$\boldsymbol{\Delta}_I=O_{\mathbb{P}}(t^{-1}),\qquad \boldsymbol{X}/t\xrightarrow{\mathbb{P}}\boldsymbol{s}^I. \tag{4.8}$$

The orthant equivalent (4.5) also gives the threshold stability

$$\mathbb{P}\{\boldsymbol{X}_I>t\boldsymbol{1}_I-\boldsymbol{\epsilon}(t)\}\sim p_I(u)\quad\text{if }\|\boldsymbol{\epsilon}(t)\|=o(t^{-1}). \tag{4.9}$$

Use q and $\mathcal{F}$ from (4.3). For $I\in\mathfrak{I}_*$, $\Sigma^{-1}\boldsymbol{s}^I$ is positive exactly on I and zero elsewhere; hence distinct sets I give distinct minimizers. By Lemma 4.1, these minimizers have disjoint open neighbourhoods U_I, $I\in\mathfrak{I}_*$, on which $x_j\le 1-\delta$ whenever $j\in A_\ell\setminus I$ and $\ell\in B_I$, for some $\delta>0$. Put $U=\bigcup_{I\in\mathfrak{I}_*}U_I$. For sufficiently small $\varepsilon>0$, coercivity gives $\vartheta>0$ such that

$$\inf\{q(\boldsymbol{x}):\boldsymbol{x}\notin U,\max_{j\in A_\ell}x_j\ge 1-\varepsilon,\ 1\le\ell\le d\}\ge\kappa+2\vartheta. \tag{4.10}$$

Otherwise a bounded sequence would converge to a minimizer outside U. On E_u, $\max_{j\in A_\ell}X_j>t-\log|A_\ell|$ for every $1\le\ell\le d$. By (4.10), on $E_u\cap\{\boldsymbol{X}/t\notin U\}$ and for large t,

$$\|\Sigma^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu})\|\ge t\sqrt{2(\kappa+2\vartheta)}-\|\Sigma^{-1/2}\boldsymbol{\mu}\|\ge t\sqrt{2(\kappa+3\vartheta/2)}.$$

Since $\|\Sigma^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu})\|^2\sim\chi^2_N$, its tail bound, with the polynomial factor absorbed into the remaining exponential gap, gives

$$\mathbb{P}\{E_u,\boldsymbol{X}/t\notin U\}\le e^{-(\kappa+\vartheta)t^2}=o(p_I(u)),\qquad I\in\mathfrak{I}_*; \tag{4.11}$$

here (4.5) gives $\log p_I(u)=-\kappa t^2+O(t+\log t)$.
Fix $I\in\mathfrak{I}_*$. On $\{\boldsymbol{X}/t\in U_I\}$, a represented group has one selected term and all other terms are at most $ue^{-\delta t}$. For $j\in I\cap A_\ell$, set $\epsilon_j(t)=-\log(1-(|A_\ell|-1)e^{-\delta t})=O(e^{-\delta t})$ and $H_I^-=\{\boldsymbol{X}_I>t\boldsymbol{1}_I-\boldsymbol{\epsilon}(t)\}$. The represented-group sum conditions lie between H_I and H_I^-, and (4.9) gives

$$\mathbb{P}\Big\{H_I^-\setminus H_I\Big\}=o(p_I(u)). \tag{4.12}$$

Together with (4.8), this yields

$$\mathbb{P}\{E_u,\boldsymbol{X}/t\in U_I\}=p_I(u)\mathbb{P}\{S_\ell>u,\ \ell\in\{1,\dots,d\}\setminus B_I\mid H_I(u)\}+o(p_I(u)). \tag{4.13}$$

Gaussian regression gives

$$(e^{X_j}/u)_{j\in I^c}=\exp\{t(\boldsymbol{s}^I_{I^c}-\boldsymbol{1}_{I^c})+\boldsymbol{d}_I+D_I\boldsymbol{\Delta}_I+\boldsymbol{\eta}_I\}. \tag{4.14}$$

By (4.6), the unrepresented groups with $\max_{j\in A_\ell}s^I_j>1$ exceed u with conditional probability tending to one, while the boundary groups have the joint limit (2.6). Since $\Gamma_I\succ 0$, the relevant residual coordinates have a Lebesgue density. The level sets of their exponential sums and maxima at one are Lebesgue-null, so the limiting margins have no atom at one. Hence the

conditional probability in (4.13) converges to $\mathbb{P}\Big\{\boldsymbol{S}^{(I)} > \mathbf{1}\Big\} > 0$. Summing over I and using (4.11) proves the sum equivalent.

For maxima, the represented-group conditions on U_I are exactly H_I for large u. The same argument, with maxima in (4.13), proves (2.7) for $\boldsymbol{T} = \boldsymbol{Q}$. Full support of the residual Gaussian law gives $0 < \mathbb{P}\Big\{\boldsymbol{Q}^{(I)} > \mathbf{1}\Big\} \le \mathbb{P}\Big\{\boldsymbol{S}^{(I)} > \mathbf{1}\Big\} \le 1$. Finally, the finite positive sums in (2.7) give

$$\lim_{u\to\infty} \frac{\log \mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\}}{(\log u)^2} = -\kappa, \qquad \boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}. \tag{4.15}$$

Expanding the exponent in (4.5) gives

$$p_I(u) \sim A_I t^{-|I|} \exp\{-\kappa t^2 + (\boldsymbol{\lambda}_I^{\mathsf{T}} \boldsymbol{\mu}_I)t\}, \qquad I \in \mathfrak{I}_*. \tag{4.16}$$

For any $I_0 \in \mathfrak{H}$, the definition of $\mathfrak{H}$ therefore implies $p_I(u)/p_{I_0}(u) \to (A_I/A_{I_0})\mathbf{1}_{\{I\in\mathfrak{H}\}}$. Consequently, (2.7) yields

$$\mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\} \sim \frac{p_{I_0}(u)}{A_{I_0}} \sum_{I\in\mathfrak{H}} A_I \mathbb{P}\Big\{\boldsymbol{T}^{(I)} > \mathbf{1}\Big\}, \qquad \boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}. \tag{4.17}$$

All weights and boundary probabilities are positive, and $\mathbb{P}\Big\{\boldsymbol{Q}^{(I)} > \mathbf{1}\Big\} \le \mathbb{P}\Big\{\boldsymbol{S}^{(I)} > \mathbf{1}\Big\}$ for every $I \in \mathfrak{H}$. Taking the ratio in (4.17) proves (1.3) with the constant (2.8), which belongs to $[1, \infty)$. □

4.3. **Grouped copulas.**

Proof of Proposition 2.3. Fix $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$ and put $z_q = \overline{\Phi}^{-1}(q) \sim \sqrt{2\log(1/q)}$. The bounds $Q_\ell \le T_\ell \le |A_\ell| Q_\ell$, together with the union bounds for the Gaussian maximum, give

$$\lim_{x\to\infty} x^{-2} \log \mathbb{P}\{\log T_\ell > x\} = -\frac{1}{2(\sigma_\ell^*)^2}, \qquad \frac{\log F^{-1}_{\ell,\boldsymbol{T}}(1-q)}{z_q} \longrightarrow \sigma_\ell^*.$$

For thresholds $h_\ell(z) = z\sigma_\ell^* + o(z)$, $\ell \in B$, we claim that

$$\lim_{z\to\infty} \frac{2}{z^2} \log \mathbb{P}\Big\{\max_{j\in A_\ell} X_j > h_\ell(z),\ \ell \in B\Big\} = -\beta_B.$$

Indeed, for every small $\varepsilon > 0$, the normalized thresholds are eventually between $(1-\varepsilon)\sigma_\ell^*$ and $(1+\varepsilon)\sigma_\ell^*$. By homogeneity, the minimum quadratic cost at these levels is $(1\pm\varepsilon)^2\beta_B$. The Gaussian norm bound used in (4.11) gives the upper rate. For the lower rate, take a minimizer $\boldsymbol{x}_*$ in (2.12) and a small ball D about $(1+\varepsilon)\boldsymbol{x}_* + \delta\mathbf{1}$, with radius less than δ. This ball is feasible for all sufficiently large z, and

$$\mathbb{P}\{\boldsymbol{X}/z \in D\} = \frac{z^N}{(2\pi)^{N/2}\sqrt{\det \Sigma}} \int_D \exp\left\{-\frac{z^2}{2}(\boldsymbol{x} - \boldsymbol{\mu}/z)^{\mathsf{T}}\Sigma^{-1}(\boldsymbol{x} - \boldsymbol{\mu}/z)\right\} d\boldsymbol{x}.$$

Bounding the integrand below by its infimum on D and then letting its radius, δ and ε tend to zero gives the lower rate; the factor z^N and the shift $\boldsymbol{\mu}/z$ do not affect it. Applying this claim at the marginal log-quantiles and at those quantiles minus $\log|A_\ell|$ proves (2.11).

For any feasible $\boldsymbol{x}$ in (2.12), select $a \in A_\ell$ and $b \in A_r$ meeting the constraints of two distinct groups $\ell, r \in B$. Cauchy–Schwarz with $\boldsymbol{w} = \boldsymbol{e}_a/\sigma_a + \boldsymbol{e}_b/\sigma_b$ gives

$$\boldsymbol{x}^{\mathsf{T}}\Sigma^{-1}\boldsymbol{x} \ge \frac{(x_a/\sigma_a + x_b/\sigma_b)^2}{2(1+\rho_{ab})} \ge \frac{2}{1+\rho_{ab}} > 1.$$

There are finitely many pairs, so $\beta_B > 1$ for $|B| \ge 2$. For $B = \{\ell\}$, the analogous bound is one and is attained by $\boldsymbol{x} = \Sigma_{I_N,\{a\}}/\sigma_a$ with $a \in A_\ell$ and $\sigma_a = \sigma_\ell^*$. If the variances are constant within two groups, the displayed pairwise bound is attained by the Gaussian regression vector with selected coordinates (σ_a, σ_b). Minimizing over the pairs proves (2.13).

For $\boldsymbol{x} \in [0,\infty)^d$, at marginal levels qx_ℓ, every pairwise intersection is bounded by $J_{\{\ell,r\},\boldsymbol{T}}(q\max_{1\le i\le d} x_i) = o(q)$ when $\max_{1\le i\le d} x_i > 0$. Bonferroni's inequalities therefore yield

$$\lim_{q\downarrow 0} \frac{1 - C_{\boldsymbol{T}}(\mathbf{1} - q\boldsymbol{x})}{q} = \sum_{\ell=1}^{d} x_\ell, \qquad \boldsymbol{x} \in [0,\infty)^d. \tag{4.18}$$

□

Proof of Corollary 2.4. Fix $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$. For $i \in \{1,2\}$, the scalar case of Theorem 2.1 gives

$$1 - F_{i,\boldsymbol{T}}(u) \sim n_i \overline{\Phi}\left(\frac{\log u - m_i}{s_i}\right). \tag{4.19}$$

Thus, with $z = \overline{\Phi}^{-1}(q)$, Mills' ratio and inversion yield

$$y_i := \frac{\log F^{-1}_{i,\boldsymbol{T}}(1-q) - m_i}{s_i} = z + \frac{\log n_i}{z} + o(z^{-1}). \tag{4.20}$$

Standardize the coordinates as $Z_j = (X_j - m_i)/s_i$, $i \in \{1,2\}$, $j \in A_i$. For $(a,b) \in A_1 \times A_2$ with correlation ρ_*, the minimum quadratic cost at thresholds one is $1/(1+\rho_*)$, and every other coordinate in these two groups has minimizing value

$$\frac{\rho_{ja} + \rho_{jb}}{1+\rho_*} < 1, \qquad j \in (A_1 \cup A_2) \setminus \{a,b\}.$$

Here the within-group correlation is strictly less than one and the cross-group correlation is at most ρ_*. Pairs with smaller correlation already have a larger cost. Hence every triple meeting both groups has a strictly larger cost than $1/(1+\rho_*)$. By finiteness, for some $\delta, \varepsilon > 0$, the probability that such a triple exceeds $(1-\delta)z$ is $O(\exp\{-[1/(1+\rho_*) + \varepsilon]z^2\})$.
Outside this event, a joint aggregate exceedance requires a unique coordinate above $(1-\delta)z$ in each group, since $T_i \le n_i \exp(m_i + s_i \max_{j\in A_i} Z_j)$. The other summands perturb the standardized thresholds by $O(e^{-s_i\delta z}) = o(z^{-1})$. The orthant equivalent (4.5) is stable under these perturbations; distinct pair intersections require a triple exceedance. Thus the joint tail is asymptotic to the sum of the pair tails with correlation ρ_*. By (4.5) and (4.20), each leading pair contributes

$$\frac{(1+\rho_*)^2}{2\pi z^2\sqrt{1-\rho_*^2}} e^{-z^2/(1+\rho_*)} (n_1 n_2)^{-1/(1+\rho_*)} (1 + o(1)).$$

Summing the n_* contributions and using $q \sim e^{-z^2/2}/(\sqrt{2\pi}z)$ and $z^2 \sim 2\log(1/q)$ proves (2.14)–(2.15). □

Proof of Corollary 2.5. Fix $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$ and write s^2 for the common variance, $\alpha_0 = 2/(1+\rho_*) > 1$, and $\widehat{C}_{\boldsymbol{T}}(u,v) = u + v - 1 + C_{\boldsymbol{T}}(1-u, 1-v)$. For $i \in \{1,2\}$, put $M_i = \max_{j\in A_i} \mu_j$ and $k_i = \#\{j \in A_i : \mu_j = M_i\}$. The scalar equivalent is $1 - F_{i,\boldsymbol{T}}(u) \sim k_i \overline{\Phi}((\log u - M_i)/s)$. With $z = \overline{\Phi}^{-1}(q)$, Mills' ratio and inversion therefore give, for each fixed $x > 0$,

$$\log F^{-1}_{i,\boldsymbol{T}}(1 - qx) = M_i + s\left(z + \frac{\log k_i - \log x}{z} + o(z^{-1})\right), \qquad i \in \{1,2\}.$$

Consequently, for $i \in \{1,2\}$ and $j \in A_i$,

$$\frac{\log F^{-1}_{i,\boldsymbol{T}}(1-q) - \mu_j}{s} = z + O(1), \qquad \frac{\log F^{-1}_{i,\boldsymbol{T}}(1-qx) - \log F^{-1}_{i,\boldsymbol{T}}(1-q)}{s} = -\frac{\log x}{z} + o(z^{-1}).$$

Put $Z_j = (X_j - \mu_j)/s$. The bound $T_i \le |A_i| \exp(M_i + s\max_{j\in A_i} Z_j)$ gives the same pair localization as above: bounded threshold shifts change the Gaussian exponents by $O(z)$, whereas the gaps for smaller correlations and triple exceedances have order z^2. Thus the joint tail is asymptotic to the sum over pairs (a,b) with $\rho_{ab} = \rho_*$. Among these pairs, those maximizing $\mu_a + \mu_b$ dominate at the linear order in z. Nevertheless, (4.5) shows that replacing q, q by qx, qy multiplies every such pair probability by $(xy)^{1/(1+\rho_*)}(1+o(1))$: the threshold

changes are $-(\log x)/z + o(z^{-1})$ and $-(\log y)/z + o(z^{-1})$, while each component of the pair's Gaussian precision vector is $z/(1+\rho_*) + O(1)$. The common multiplier therefore survives the finite sum, irrespective of the pairs' relative weights. Consequently,

$$\frac{\widehat{C}_{\boldsymbol{T}}(qx, qy)}{\widehat{C}_{\boldsymbol{T}}(q, q)} \longrightarrow (xy)^{\alpha_0/2}, \qquad q \downarrow 0, \quad x, y > 0.$$

Taking $x = y = c$ shows that the diagonal tail is regularly varying at zero with index α_0. Choose b_0 so that $r\widehat{C}_{\boldsymbol{T}}(1/b_0(r), 1/b_0(r)) \to 1$ as $r \to \infty$. Then b_0 is regularly varying with index $1/\alpha_0$ and $b_0(r)/r \to 0$, while the ratio limit gives

$$r\mathbb{P}\{W_{1,\boldsymbol{T}} > b_0(r)x,\ W_{2,\boldsymbol{T}} > b_0(r)y\} \longrightarrow \nu_0((x,\infty)\times(y,\infty)) = (xy)^{-\alpha_0/2}, \qquad x, y > 0.$$

For each $\varepsilon > 0$, the limsup of the normalized mass where $\min(w_1, w_2) > \varepsilon$ and $\max(w_1, w_2) > M$ is at most $2(M\varepsilon)^{-\alpha_0/2}$, which tends to zero. On each compact set bounded away from the axes, the rectangle limits are convergence-determining. The preceding bound controls the mass escaping to infinity and proves M_0-convergence on $(0,\infty)^2$, that is, convergence against bounded continuous functions vanishing on a neighbourhood of the axes. The limit is homogeneous of order $-\alpha_0$. Finally, standard Pareto margins and (4.18) give ordinary regular variation with index one and mass on the axes. □

Proof of Remark 2.6. Put $u = F^{-1}_{1,\boldsymbol{Q}}(1-q)$ and $t = \log u$. For suitable $\sigma > 0$, $M \in \mathbb{R}$ and a positive integer k, the Gaussian maximum in the first group satisfies

$$\mathbb{P}\{Q_1 > e^v\} \sim k\overline{\Phi}((v-M)/\sigma) = e^{\psi(v)}, \qquad \psi'(v) \sim -v/\sigma^2.$$

By the hypothesis and the scalar case of Theorem 2.1, this equivalent holds for every T_ℓ, $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$. Hence, writing $u_{\ell,\boldsymbol{T}} = F^{-1}_{\ell,\boldsymbol{T}}(1-q)$, we have $\psi(\log u_{\ell,\boldsymbol{T}}) - \psi(t) \to 0$. Mills' ratio and the mean value theorem give $\log u_{\ell,\boldsymbol{T}} = t + o(t^{-1})$ for all $2d$ quantiles. Let u_- and u_+ be their minimum and maximum. Then

$$\mathbb{P}\{\boldsymbol{T} > u_+\mathbf{1}\} \le J_{B,\boldsymbol{T}}(q) \le \mathbb{P}\{\boldsymbol{T} > u_-\mathbf{1}\}, \qquad B = \{1, \dots, d\}.$$

Since $\log u_\pm = t + o(t^{-1})$, (2.7) and the stability (4.9) show that both bounds are asymptotic to $\mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\}$. Taking the ratio for sums and maxima and using (1.3) proves (2.16). □

Proof of Corollary 2.7. Fix a group ℓ and let $p \downarrow 0$. Put

$$A_p = \{Q_\ell > F^{-1}_{\ell,\boldsymbol{Q}}(1-p)\},$$
$$B_p = \{S_\ell > F^{-1}_{\ell,\boldsymbol{S}}(1-p)\},$$
$$D_p = \{S_\ell > F^{-1}_{\ell,\boldsymbol{Q}}(1-p)\}.$$

Since $Q_\ell \le S_\ell$, their quantiles are ordered and $A_p, B_p \subset D_p$. Continuity gives $\mathbb{P}\{A_p\} = \mathbb{P}\{B_p\} = p$, while the scalar case of Theorem 2.1 gives $\mathbb{P}\{D_p\} = p + o(p)$. Hence

$$\mathbb{P}\{A_p \triangle B_p\} \le 2(\mathbb{P}\{D_p\} - p) = o(p).$$

For fixed $\boldsymbol{x}, \boldsymbol{y}$, we can therefore replace each sum exceedance at marginal probability level qx_ℓ by the corresponding maximum exceedance, with total error $o(q)$; zero coordinates are omitted. With $z_\ell = \max(x_\ell, y_\ell)$ this yields

$$C_{\boldsymbol{S},\boldsymbol{Q}}(\mathbf{1} - q\boldsymbol{x}, \mathbf{1} - q\boldsymbol{y}) = C_{\boldsymbol{Q}}(\mathbf{1} - q\boldsymbol{z}) + o(q).$$

Equation (4.18) proves (2.17). Its first-order expansion also gives

$$C_{\boldsymbol{S},\boldsymbol{Q}}(\boldsymbol{u}^{1/n}, \boldsymbol{v}^{1/n})^n \longrightarrow \exp\left\{-\sum_{\ell=1}^{d} \max(-\log u_\ell, -\log v_\ell)\right\} = \prod_{\ell=1}^{d} \min(u_\ell, v_\ell)$$

for $\boldsymbol{u}, \boldsymbol{v} \in (0,1]^d$, proving (2.18). □

4.4. **Examples.**

Proof of Examples 3.1 and 3.2 and Remark 3.3. In Example 3.1, the only admissible set is $I = \{1\}$, with $\boldsymbol{s}^I = \mathbf{1}_3$ and $\kappa = 1/2$. Indeed, either other singleton has first slope $1/2$, whereas a pair containing coordinate 1 has a zero multiplier. The boundary variables have law (Y_1, Y_2), so their maximum exceeds zero with probability $1 - \Phi(0)^2 = 3/4$. Formula (2.8) gives the stated constant. Its strict bounds follow because $\{\max(Y_1, Y_2) > 0\}$ is a proper subset, up to positive probability, of $\{e^{Y_1} + e^{Y_2} > 1\}$, whose complement also has positive probability.
For distinct coordinate–time pairs in Example 3.2,

$$\operatorname{Cov}(X_i(k), X_j(l)) = \tfrac{1}{4} + \tfrac{1}{2}\cos((k-l)\pi/3).$$

The independent noise contributes $I/4$ to each finite sampled covariance matrix, proving positive definiteness. Singletons are infeasible; pairs at adjacent times have slope $1/3$ at the remaining time and are also infeasible. Each one-per-time triple has multiplier vector $(1, 0, 1)^{\mathsf{T}}$. Thus $\mathfrak{I}_*$ consists of the four pairs selecting one coordinate at time 0 and one at time 2. Each has $\kappa = 1$, and its only boundary group is at time 1. Its two conditional variables, denoted by Y_1, Y_2, have mean zero and covariance

$$\begin{pmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{pmatrix}.$$

Consequently, $\mathbb{P}\{\max(Y_1, Y_2) > 0\} = 1 - (1/4 + \arcsin(1/2)/(2\pi)) = 2/3$, and (2.8) yields $C_* = \frac{3}{2}\mathbb{P}\left\{e^{Y_1} + e^{Y_2} > 1\right\} \in (1, 3/2)$. For any two time groups, Remark 2.2 and Corollary 2.4 give the remaining claims.
In Remark 3.3, every admissible set is a cross-group pair with covariance c, multiplier vector $(v + c)^{-1}\mathbf{1}_2$ and cost $(v + c)^{-1}$. Hence exactly the n_* pairs with covariance c_* are dominant. They have identical Gaussian laws and no boundary groups, so Theorem 2.1 gives (3.5); Corollary 2.4 gives the asserted copula equivalent. □

Proof of Example 3.4. For a selector I with $|I| = k$, the equicorrelated matrix gives

$$\boldsymbol{\lambda}_I = \frac{\mathbf{1}_I}{\sigma^2 D_k}, \qquad s_j^I = \frac{k\rho}{D_k} < 1 \quad (j \notin I),$$

since $D_k > 0$ and $D_k - k\rho = 1 - \rho > 0$. Thus an unrepresented group cannot satisfy (2.2). The admissible sets are exactly the selectors containing one coordinate from each group, and they all have cost $d/(2\sigma^2 D_d)$ and no boundary groups. Their selected Gaussian vectors have identical laws. Theorem 2.1 and (4.5), using

$$\det\left[(1-\rho)\mathrm{Id}_d + \rho\mathbf{1}\mathbf{1}^{\mathsf{T}}\right] = (1-\rho)^{d-1} D_d,$$

prove $C_* = 1$ and (3.6). Applying the same calculation to the groups in B gives the minimum standardized quadratic cost b/D_b in (2.12), proving (3.7). □

4.5. **Signed sums.**

Proof of Proposition 3.6. Write $q(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\Sigma^{-1}\boldsymbol{x}$ and $\kappa_\pm = \min_{\mathcal{F}_\pm} q$. The set $\mathcal{F}_\pm$ is the union, over $p_\ell \in P_\ell$, of the polyhedra

$$\{\boldsymbol{x} : x_{p_\ell} \geq 1 \ (1 \leq \ell \leq d), \quad x_{p_\ell} \geq x_k \ (1 \leq \ell \leq d, \ k \in N_\ell)\}.$$

Each has a unique quadratic minimizer, so $\mathcal{S}_\pm$ is finite and nonempty. Fix $\boldsymbol{s} \in \mathcal{S}_\pm$ and select $s_{p_\ell} = h_\ell$. The optimality conditions give

$$\boldsymbol{g} = \sum_{1 \leq \ell \leq d} \left[\beta_\ell \boldsymbol{e}_{p_\ell} + \sum_{k \in N_\ell} \nu_{\ell k}(\boldsymbol{e}_{p_\ell} - \boldsymbol{e}_k)\right], \qquad \beta_\ell, \nu_{\ell k} \geq 0, \tag{4.21}$$

with $\beta_\ell = \sum_{j\in A_\ell} g_j$ and complementary slackness. Put $B = \{\ell : \boldsymbol{g}_{A_\ell} \neq \mathbf{0}\}$. For $\ell \in B$, $g_{p_\ell} > 0$, and p_ℓ is the unique positive maximizer: choosing another maximizer in its place would force $g_{p_\ell} = 0$ in (4.21). Define

$$K_\ell = \{k \in N_\ell : \nu_{\ell k} > 0\}, \qquad a_\ell = g_{p_\ell} = \beta_\ell + \sum_{k\in K_\ell} \nu_{\ell k}.$$

Complementary slackness gives $s_k = h_\ell$ for $k \in K_\ell$ and $\beta_\ell(h_\ell - 1) = 0$. For $j \in P_\ell \setminus \{p_\ell\}$, uniqueness of the positive maximizer gives $s_j < h_\ell$. For $k \in N_\ell \setminus K_\ell$, (4.21) gives $g_k = -\nu_{\ell k} = 0$, so the second implication in (3.14) gives $s_k < h_\ell$. Thus all omitted coordinates in an active group have slope strictly below h_ℓ. With

$$B^1 = \{\ell \in B : h_\ell = 1\}, \qquad B^> = \{\ell \in B : h_\ell > 1\},$$

the first implication in (3.14) gives $\beta_\ell > 0$ on B^1, whereas complementary slackness gives $a_\ell = \sum_{k\in K_\ell} \nu_{\ell k} > 0$ on $B^>$. Moreover, $2\kappa_\pm = \boldsymbol{g}^\mathsf{T}\boldsymbol{s} = \sum_{\ell\in B} \beta_\ell > 0$, so $B^1 \neq \varnothing$.
Let $L\boldsymbol{X}$ collect $(X_{p_\ell}, (X_k)_{k\in K_\ell})$ for $\ell \in B^1$ and $(X_k - X_{p_\ell})_{k\in K_\ell}$ for $\ell \in B^>$. Its n rows are linearly independent. Put

$$K = L\Sigma L^\mathsf{T}, \qquad \boldsymbol{m} = L\boldsymbol{\mu}, \qquad \boldsymbol{c} = L\boldsymbol{s}, \qquad \boldsymbol{v} = K^{-1}\boldsymbol{c}.$$

The corresponding subvectors are

$$(\boldsymbol{c}_\ell, \boldsymbol{v}_\ell) = \begin{cases} ((1, \mathbf{1}), (a_\ell, (-\nu_{\ell k})_{k\in K_\ell})), & \ell \in B^1, \\ (\mathbf{0}, (-\nu_{\ell k})_{k\in K_\ell}), & \ell \in B^>. \end{cases}$$

Indeed, (4.21) gives $\boldsymbol{g} = L^\mathsf{T}\boldsymbol{v}$, and hence

$$\boldsymbol{s} = \Sigma L^\mathsf{T} K^{-1}\boldsymbol{c}, \qquad \tfrac{1}{2}\boldsymbol{c}^\mathsf{T} K^{-1}\boldsymbol{c} = \kappa_\pm. \tag{4.22}$$

Let $H_s(u)$ be the event

$$\begin{cases} e^{X_{p_\ell}} - \sum_{k\in K_\ell} e^{X_k} > u, & \ell \in B^1, \\ \sum_{k\in K_\ell} e^{X_k - X_{p_\ell}} < 1, & \ell \in B^>. \end{cases}$$

Within each group factor below, write $p = p_\ell$ and let every index k range over K_ℓ; y_{k-p} denotes the entry corresponding to $X_k - X_{p_\ell}$. Writing $L\boldsymbol{X} = t\boldsymbol{c} + \boldsymbol{y}$ gives a fixed product domain $\mathcal{D}$: its factors are $e^{y_p} - \sum_k e^{y_k} > 1$ on B^1 and $\sum_k e^{y_{k-p}} < 1$ on $B^>$. Therefore

$$\mathbb{P}\{H_s(u)\} = \frac{e^{-\kappa_\pm t^2 + t\boldsymbol{v}^\mathsf{T}\boldsymbol{m}}}{(2\pi)^{n/2}\sqrt{\det K}} \int_{\mathcal{D}} e^{-t\boldsymbol{v}^\mathsf{T}\boldsymbol{y}} e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{m})^\mathsf{T} K^{-1}(\boldsymbol{y}-\boldsymbol{m})} d\boldsymbol{y}. \tag{4.23}$$

The first factor integrates to

$$\int_{\mathcal{D}} e^{-t\boldsymbol{v}^\mathsf{T}\boldsymbol{y}} d\boldsymbol{y} = \prod_{\ell\in B^1} \frac{\Gamma(t\beta_\ell)\prod_{k\in K_\ell}\Gamma(t\nu_{\ell k})}{\Gamma(ta_\ell + 1)} \prod_{\ell\in B^>} \frac{\prod_{k\in K_\ell}\Gamma(t\nu_{\ell k})}{\Gamma(ta_\ell + 1)}. \tag{4.24}$$

To see both this identity and the associated concentration, substitute $z_k = e^{y_k}$ on B^1 and integrate y_p first. Under the resulting normalized measure, $z_k = G_k/G_0$, with independent unit-scale gamma variables of shapes $t\nu_{\ell k}$ and $t\beta_\ell$, while $y_p = \log(1 + \sum_k z_k) + E/(ta_\ell)$ with E standard exponential. On $B^>$, $z_k = e^{y_{k-p}}$ and their slack have Dirichlet parameters $((t\nu_{\ell k})_k, 1)$. Thus the normalized measure in (4.24) concentrates at $\boldsymbol{b}$, whose group entries are

$$\boldsymbol{b}_\ell = \begin{cases} (\log(a_\ell/\beta_\ell), (\log(\nu_{\ell k}/\beta_\ell))_k), & \ell \in B^1, \\ (\log(\nu_{\ell k}/a_\ell))_k, & \ell \in B^>. \end{cases}$$

The Gaussian factor in (4.23) is bounded, continuous and positive at $\boldsymbol{b}$. Stirling's formula in (4.24) therefore gives

$$\mathbb{P}\{H_s(u)\} \sim A_s t^{-\rho_s} e^{-\kappa_\pm t^2 - \alpha_s t}, \qquad \rho_s = (n + |B|)/2, \quad \alpha_s = \boldsymbol{v}^\mathsf{T}(\boldsymbol{b} - \boldsymbol{m}), \quad A_s > 0. \tag{4.25}$$

More explicitly,

$$A_s = \frac{\exp\Big(-\frac{1}{2}(\boldsymbol{b}-\boldsymbol{m})^\top K^{-1}(\boldsymbol{b}-\boldsymbol{m})\Big)}{(2\pi)^{|B|/2}\sqrt{\det K}} \prod_{\ell\in B}\left(a_\ell \prod_{k\in K_\ell}\nu_{\ell k}\right)^{-1/2} \prod_{\ell\in B^1}\beta_\ell^{-1/2},$$

where empty products equal one. The same concentration argument gives $L\boldsymbol{X} - t\boldsymbol{c} \to \boldsymbol{b}$ in conditional probability. Gaussian regression and (4.22) now imply

$$\mathrm{Law}(\boldsymbol{X} - t\boldsymbol{s} \mid H_s(u)) \xrightarrow{\mathrm{w}} \mathrm{Law}(\boldsymbol{W}_s), \qquad \mathrm{Law}(\boldsymbol{W}_s) = \mathrm{Law}(\boldsymbol{X} \mid L\boldsymbol{X} = \boldsymbol{b}). \tag{4.26}$$

In particular, $\boldsymbol{X}/t \to \boldsymbol{s}$ under this conditioning.
Choose small disjoint neighbourhoods U_s of the points in $\mathcal{S}_\pm$. A signed exceedance implies

$$\max_{j\in P_\ell} X_j \ge t - \log|P_\ell|, \qquad \max_{j\in P_\ell} X_j \ge \max_{k\in N_\ell} X_k - \log|P_\ell|.$$

Coercivity gives a strict quadratic-cost gap outside $\bigcup_s U_s$. The Gaussian norm bound used in (4.11), together with (4.25), makes this contribution $o(\mathbb{P}\{H_s(u)\})$ for every s.
Inside U_s, omitted terms in a group of B^1 have total absolute value $O(ue^{-\delta t})$ for some $\delta > 0$. The full sum and adjusted-maximum conditions are therefore bounded by the corresponding condition in $H_s(u)$ at thresholds $u(1 \pm O(e^{-\delta t}))$. For $B^>$, division by $e^{X_{p_\ell}}$ gives the bounds $\sum_{k\in K_\ell} e^{X_k - X_{p_\ell}} < 1 \pm O(e^{-\delta t})$, since both the omitted terms and $u/e^{X_{p_\ell}}$ are exponentially small. Choose a deterministic $\varepsilon_t = Ce^{-\delta t}$ that bounds all these perturbations. The lower and upper domains are $\mathcal{D} + \boldsymbol{\tau}_t^-$ and $\mathcal{D} + \boldsymbol{\tau}_t^+$, respectively: on a B^1 block, all entries of $\boldsymbol{\tau}_t^-$ and $\boldsymbol{\tau}_t^+$ are $\log(1+\varepsilon_t)$ and $\log(1-\varepsilon_t)$; on a $B^>$ block, these two shifts are interchanged. Indeed, a common shift of the log-variables multiplies the corresponding exponential sum or difference by the exponential of that shift. Both translations are $O(e^{-\delta t}) = o(t^{-1})$. After translating back to $\mathcal{D}$ in (4.23), the extra factor $e^{-t\boldsymbol{v}^\top\boldsymbol{\tau}_t^\pm}$ tends to one and the Gaussian factor has the same limit at $\boldsymbol{b}$. Both bounds also satisfy $\boldsymbol{X}/t \to \boldsymbol{s}$ conditionally. Their probability difference is therefore $o(\mathbb{P}\{H_s(u)\})$, also after restriction to U_s.
For $\ell \in \{1,\dots,d\} \setminus B$, put $J_\ell^+ = \{j \in P_\ell : s_j = h_\ell\}$ and $J_\ell^- = \{k \in N_\ell : s_k = h_\ell\}$. The boundary probabilities obtained from (4.26) are

$$C_s = \mathbb{P}\left\{\sum_{j\in J_\ell^+} e^{W_{s,j}} - \sum_{k\in J_\ell^-} e^{W_{s,k}} > \mathbf{1}_{\{h_\ell=1\}}, \quad \ell \in \{1,\dots,d\}\setminus B\right\},$$

$$D_s = \mathbb{P}\left\{\max_{j\in J_\ell^+} e^{W_{s,j}} - \sum_{k\in J_\ell^-} e^{W_{s,k}} > \mathbf{1}_{\{h_\ell=1\}}, \quad \ell \in \{1,\dots,d\}\setminus B\right\}.$$

If $B = \{1,\dots,d\}$, set $C_s = D_s = 1$. The coordinates of $\boldsymbol{W}_s$ in these groups have positive definite covariance, because L only involves active groups. Their exceedance boundaries are null: condition on all but one positive coordinate for sums, or all but one negative coordinate for adjusted maxima; without negative terms, use continuity of the Gaussian maximum. Full support and event inclusion give $0 < D_s \le C_s \le 1$. Thus localization and (4.26) yield

$$\mathbb{P}\Big\{\widetilde{\boldsymbol{S}} > u\mathbf{1}\Big\} \sim \sum_{s\in\mathcal{S}_\pm} C_s\mathbb{P}\{H_s(u)\}, \qquad \mathbb{P}\Big\{\widetilde{\boldsymbol{Q}} > u\mathbf{1}\Big\} \sim \sum_{s\in\mathcal{S}_\pm} D_s\mathbb{P}\{H_s(u)\}.$$

Let $\mathfrak{H}_\pm$ consist of the minimizers in $\mathcal{S}_\pm$ with smallest α_s and, among these, smallest ρ_s. By (4.25), only these terms contribute to the two finite sums, so

$$C_\pm = \frac{\sum_{s\in\mathfrak{H}_\pm} A_sC_s}{\sum_{s\in\mathfrak{H}_\pm} A_sD_s} \in [1,\infty).$$

For $d = 1$, $B = \{1\}$, so $C_s = D_s = 1$ for every minimizer and $C_\pm = 1$. Without negative terms, $B^> = \varnothing$, $K_\ell = \varnothing$, $a_\ell = \beta_\ell$ and $\boldsymbol{b} = \mathbf{0}$. Then L selects $I = \{p_\ell : \ell \in B\}$, and the above quantities become $A_s = A_I$, $\alpha_s = -\boldsymbol{\lambda}_I^\mathsf{T}\boldsymbol{\mu}_I$ and $\rho_s = |I|$. The correspondence $\boldsymbol{s} = \boldsymbol{s}^I$ identifies $\mathfrak{H}_\pm$ with $\mathfrak{H}$, and the displayed ratio reduces to (2.8).

□

4.6. **Aggregation and common excess.**

Proof of Remark 3.5. Put $t = \log u$ and $f(u) = \mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\}$. By (4.16) and (4.17), for constants $K, m > 0$ and $\gamma \in \mathbb{R}$,

$$f(e^t) \sim \Psi(t) := Kt^{-m}e^{-\kappa t^2+\gamma t}.$$

Since $\{S_\wedge > v\} = \{\boldsymbol{S} > v\mathbf{1}\}$, for $x \geq 0$,

$$\mathbb{P}\Big\{\frac{2\kappa t}{u}(S_\wedge - u) > x \mid \boldsymbol{S} > u\mathbf{1}\Big\} = \frac{f(u(1 + x/(2\kappa t)))}{f(u)} \longrightarrow e^{-x}.$$

This proves (3.10) and gives $S_\wedge/u \to 1$ conditionally. Tonelli's theorem also gives

$$\mathbb{E}\{S_\wedge - u \mid \boldsymbol{S} > u\mathbf{1}\} = \frac{\int_t^\infty e^s f(e^s)\,ds}{f(e^t)}.$$

The relation $f(e^s)/\Psi(s) \to 1$ is uniform over $s \geq t$ as $t \to \infty$. Substitution $s = t + z/t$ yields

$$\frac{t\int_t^\infty e^s\Psi(s)\,ds}{e^t\Psi(t)} = \int_0^\infty (1 + z/t^2)^{-m}e^{-2\kappa z-\kappa z^2/t^2+(\gamma+1)z/t}\,dz \longrightarrow \frac{1}{2\kappa},$$

since the integrand is bounded by $e^{-\kappa z}$ for large t. This proves (3.11). For signed sums, (4.25) and the finite mixtures in the proof of Proposition 3.6 give the same form of Ψ, with $\kappa_\pm$ replacing κ, $m = \rho_s > 0$ and $\gamma = -\alpha_s$ on the dominant set. The same argument proves the signed extension.

For Remark 3.5, write $Q_\wedge = \min_\ell Q_\ell$ and $E_u = \{\boldsymbol{S} > u\mathbf{1}\}$. The localization in (4.13), retaining the joint boundary limit (2.6), gives, for every $r > 1$,

$$\mathbb{P}\{E_u, Q_\wedge > u/r\} \sim \sum_{I\in\mathfrak{I}_*} p_I(u)\mathbb{P}\Big\{\boldsymbol{S}^{(I)} > \mathbf{1}, \boldsymbol{Q}^{(I)} > r^{-1}\mathbf{1}\Big\}.$$

Indeed, represented groups exceed u apart from (4.12), and unrepresented groups with largest slope greater than one do so with conditional probability tending to one. Dividing by (4.17) gives the expression in (3.9) for the limiting conditional cdf of $u/Q_\wedge$ at $r > 1$. It is continuous there and tends to one as $r \to \infty$. For $0 < r < 1$, the corresponding probability is at most $\mathbb{P}\{\boldsymbol{Q} > (u/r)\mathbf{1}\}/f(u) \to 0$ by (4.16) and (4.17). Thus the limit is supported on $[1, \infty)$, and Slutsky's theorem, using $S_\wedge/u \to 1$, gives the same limit for R_u. Letting $r \downarrow 1$ in the limiting cdf, atomlessness of the boundary maxima and $\boldsymbol{Q}^{(I)} \leq \boldsymbol{S}^{(I)}$ give

$$\mathbb{P}\{\mathcal{R} = 1\} = \frac{\sum_{I\in\mathfrak{H}} A_I\mathbb{P}\Big\{\boldsymbol{Q}^{(I)} > \mathbf{1}\Big\}}{\sum_{I\in\mathfrak{H}} A_I\mathbb{P}\{\boldsymbol{S}^{(I)} > \mathbf{1}\}} = C_*^{-1}.$$

This proves (3.8)–(3.9) and the stated equivalence. □

4.7. **Conditional risk measures.**

Proof of (3.16) and Remark 3.7. Writing $f_{\boldsymbol{\mu}}$ for the Gaussian density, completion of the square gives

$$e^{x_a}f_{\boldsymbol{\mu}}(\boldsymbol{x}) = e^{\mu_a+\Sigma_{aa}/2}f_{\boldsymbol{\mu}+\Sigma\boldsymbol{e}_a}(\boldsymbol{x}).$$

Integrating over $\mathcal{E}_B(u)$, summing over $a \in A_\ell$ and conditioning proves (3.16).

For the remainder, use Remark 3.3 and write $t = \log u$ and $c_{kl} = \mathrm{Cov}(X_i(k), X_j(l))$. Applying the scalar case of Theorem 2.1 under each tilted mean gives

$$\mathrm{MES}_i(u) \sim \frac{e^{\mu_i+v/2}}{n_j\overline{\Phi}((t-\mu_j)/\sqrt{v})}\sum_{k\in T_i}\sum_{l\in T_j}\overline{\Phi}\Big(\frac{t-\mu_j-c_{kl}}{\sqrt{v}}\Big).$$

Mills' ratio shows that the n_* pairs with $c_{kl} = c_*$ dominate and gives (3.17).
To prove (3.18), first consider a leading Gaussian pair $\boldsymbol{Y} = (Y_1, Y_2)^\mathsf{T}$ with mean $(\mu_i, \mu_j)^\mathsf{T}$ and covariance $K = \left(\begin{smallmatrix} v & c_* \\ c_* & v \end{smallmatrix}\right)$. Put $\lambda = (v + c_*)^{-1} > 0$, $p(u) = \mathbb{P}\{\boldsymbol{Y} > t\mathbf{1}\}$ and $G(u) = \mathbb{E}\Big\{(e^{Y_1} - u)_+ \mathbf{1}_{\{Y_2 > t\}}\Big\}$. In the Gaussian density substitute $\boldsymbol{Y} = t\mathbf{1} + (h,k)^\mathsf{T}/t$, with $h, k > 0$. If f denotes this density and $\boldsymbol{g} = K^{-1}(t\mathbf{1} - (\mu_i, \mu_j)^\mathsf{T})$, then

$$\frac{f(t\mathbf{1} + (h,k)^\mathsf{T}/t)}{f(t\mathbf{1})} = \exp\left\{ -\frac{g_1 h + g_2 k}{t} - \frac{(h,k) K^{-1} (h,k)^\mathsf{T}}{2t^2} \right\}, \qquad \frac{g_1}{t}, \frac{g_2}{t} \longrightarrow \lambda.$$

Moreover, $t(e^{h/t} - 1) \le h e^{h/t}$, so for large t the integrand with this additional factor is bounded by $h e^{-\lambda h/4 - \lambda k/2}$. Dominated convergence in the numerator and denominator therefore yields

$$G(u) \sim \frac{u}{t} p(u) \frac{\int_0^\infty \int_0^\infty h e^{-\lambda(h+k)} \, dh \, dk}{\int_0^\infty \int_0^\infty e^{-\lambda(h+k)} \, dh \, dk} = (v + c_*) \frac{u}{t} p(u). \tag{4.27}$$

Now apply the localization proof of Theorem 2.1 to the two groups. Their dominant configurations are the n_* pairs with covariance c_*, and their common quadratic cost is $\kappa = (v + c_*)^{-1}$. In disjoint neighbourhoods of these configurations, each group has one selected coordinate and the omitted summands total at most $\varepsilon_u = Cue^{-\delta t}$, for fixed $C, \delta > 0$. The weighted contributions outside these neighbourhoods are negligible: bound the payoff by S_i and apply the density-tilting identity above to each summand. Each tilted Gaussian has the same covariance and hence the same strict quadratic-cost gap as in (4.10); the Gaussian norm bound gives $O(e^{-(\kappa+\eta)t^2})$, for some $\eta > 0$, which is $o(up(u)/t)$. The same argument controls the selected-pair payoff outside its own neighbourhood.
Within each neighbourhood, the aggregate payoff with the stress indicator is bounded below by the selected-pair payoff at u and above by that payoff at $u - \varepsilon_u$, with both pair thresholds shifted together. Since $\log(u - \varepsilon_u) = t + o(t^{-1})$, (4.9) and (4.27) give $G(u - \varepsilon_u) \sim G(u)$. Consequently, as $u \to \infty$

$$\mathbb{E}\Big\{(S_i - u)_+ \mathbf{1}_{\{S_j > u\}}\Big\} \sim n_* G(u) \sim (v + c_*) \frac{u}{t} \mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\},$$

where the last step uses (3.5). Dividing by $\mathbb{P}\{S_j > u\}$ proves (3.18). □

4.8. **Log-elliptical sums.**

Proof of Proposition 3.8. Use (3.19), and retain the deterministic quantities in (4.1) and the regression coefficients in (2.4). For $I \in \mathfrak{I}_*$, put $H_I(u) = \{\boldsymbol{X}_I > t\mathbf{1}_I\}$, $v = \sqrt{2\kappa}$ and $h_I(t) = \sqrt{r_I(t)/w(r_I(t))}$. The elliptical orthant and conditional limit results in [19, Theorems 3.1, 3.4 and Corollary 5.3] give

$$p_I(u) \sim \frac{\mathcal{B}_N(r_I(t)) [r_I(t)/w(r_I(t))]^{|I|/2}}{(2\pi)^{|I|/2} \sqrt{\det \Sigma_{II}} \prod_{j \in I} g_{I,j}(t)}, \tag{4.28}$$

where

$$\mathcal{B}_N(r) = 2^{N/2-1} \Gamma(N/2) [rw(r)]^{1-N/2} \overline{F}(r),$$

and, with $\boldsymbol{\Delta}_I, \boldsymbol{\eta}_I$ defined algebraically by (4.2),

$$\operatorname{Law}\left(w(r_I(t)) \boldsymbol{\Delta}_I, \frac{\boldsymbol{\eta}_I}{h_I(t)} \mid H_I(u) \right) \xrightarrow[u \to \infty]{\mathrm{w}} \left(\bigotimes_{j \in I} \operatorname{Exp}(\lambda_{I,j}/v) \right) \otimes \mathcal{N}_{N-|I|}(\mathbf{0}, \Gamma_I). \tag{4.29}$$

Empty residuals are omitted, and no radial density is required. To check the moving-threshold hypotheses, write $\boldsymbol{z}_I(t) = t\mathbf{1}_I - \boldsymbol{\mu}_I$. The selected subvector has all its coordinates active for large t, since

$$\Sigma_{II}^{-1} \boldsymbol{z}_I(t) = t\boldsymbol{\lambda}_I - \Sigma_{II}^{-1} \boldsymbol{\mu}_I > \mathbf{0}_I, \qquad \frac{\boldsymbol{z}_I(t)}{r_I(t)} \longrightarrow \frac{\mathbf{1}_I}{v}.$$

Thus the radius and scale in Theorem 3.1 of [19] are exactly $r_I(t)$ and $w(r_I(t))$; its limiting active multipliers are $\boldsymbol{\lambda}_I / v > \mathbf{0}_I$. Applying the projection identity used in its Lemma 3.5 gives

(4.28) in terms of the original N-dimensional radius R. For the conditional result, extend the centred threshold to I^c by $D_I \boldsymbol{z}_I(t)$. Its minimal active set is I, and the scaled residual threshold in that paper's condition (3.11) is zero. Corollary 5.3 therefore applies. Replacing the deterministic regression centre $\boldsymbol{\mu}_{I^c} + D_I \boldsymbol{z}_I(t)$ by $\boldsymbol{\mu}_{I^c} + D_I(\boldsymbol{X}_I - \boldsymbol{\mu}_I)$ changes the normalized residual by $O_{\mathbb{P}}((r_I(t)w(r_I(t)))^{-1/2}) = o_{\mathbb{P}}(1)$, which proves (4.29) with (4.2). For $N = 1$, (4.28) reduces directly to $p_I(u) = \overline{F}(r_I(t))/2$, and the conditional limit is (3.20).
Both cases have $w \in \mathrm{RV}_a$ for some $a > -1$, with $a = 1$ in part (i). The Gumbel representation theorem [28, Chapter 1] gives, for large r_0,

$$\overline{F}(r) = c(r) \exp\left\{ - \int_{r_0}^{r} \widetilde{w}(s)\, ds \right\}, \qquad c(r) \longrightarrow c_0 \in (0, \infty), \qquad \widetilde{w}(r) \sim w(r). \tag{4.30}$$

Consequently, Karamata's theorem gives

$$H(r) := -\log \overline{F}(r) \sim \int_{r_0}^{r} w(s)\, ds \sim \frac{r w(r)}{a+1} \in \mathrm{RV}_{a+1}\,. \tag{4.31}$$

Use the disjoint neighbourhoods and strict quadratic-cost gap from Lemma 4.1 and (4.10). Outside their union, a sum exceedance requires $R \geq (v + \varepsilon)t - O(1)$ for some $\varepsilon > 0$. The last display and (4.28) make this contribution $o(p_I(u))$ for every $I \in \mathfrak{I}_*$. Also, $r_I(t)w(r_I(t)) \to \infty$, so (4.29) gives $\boldsymbol{X}/t \to \boldsymbol{s}^I$ conditionally on $H_I(u)$.
Inside a dominant neighbourhood, the unused terms in a represented group perturb its selected log-threshold by $O(e^{-\delta t})$. Since $w(r_I(t))e^{-\delta t} \to 0$, the elliptical orthant equivalent is stable under this perturbation. Thus the local reduction (4.13) holds for sums and maxima under the elliptical law. Unrepresented groups with $\max_{j \in A_\ell} s_j^I > 1$ exceed u with conditional probability tending to one.
In part (i), $h_I(t) \to \tau$ and $\boldsymbol{\Delta}_I \to \boldsymbol{0}_I$ conditionally on $H_I(u)$. The identity (4.14) and (4.29) give

$$\mathrm{Law}\left(\frac{\boldsymbol{S}_{\mathcal{C}_I}}{u}, \frac{\boldsymbol{Q}_{\mathcal{C}_I}}{u} \mid H_I(u) \right) \xrightarrow[u\to\infty]{\mathrm{w}} \mathrm{Law}(\boldsymbol{S}_\tau^{(I)}, \boldsymbol{Q}_\tau^{(I)}).$$

The limiting margins are atomless at one. Summing the local contributions proves (3.21). For the ratio, write

$$r_I(t)^2 = 2\kappa t^2 - 2(\boldsymbol{\lambda}_I^{\mathsf{T}} \boldsymbol{\mu}_I)t + \boldsymbol{\mu}_I^{\mathsf{T}} \Sigma_{II}^{-1} \boldsymbol{\mu}_I.$$

Write $\gamma_I = \boldsymbol{\lambda}_I^{\mathsf{T}} \boldsymbol{\mu}_I$ and $\zeta_I = \boldsymbol{\mu}_I^{\mathsf{T}} \Sigma_{II}^{-1} \boldsymbol{\mu}_I$. Fix $I_0 \in \mathfrak{H}$. If $\gamma_I < \gamma_{I_0}$, then $r_I(t) - r_{I_0}(t) \to (\gamma_{I_0} - \gamma_I)/v > 0$. By (4.30) and $w(r) \sim r/\tau^2$, for each fixed $b > 0$,

$$\log \frac{\overline{F}(r+b)}{\overline{F}(r)} = -\frac{br}{\tau^2}(1 + o(1)).$$

Thus the contribution with the smaller score is negligible even after multiplication by any fixed power of t. If the scores are tied,

$$r_I(t) - r_{I_0}(t) = \frac{\zeta_I - \zeta_{I_0}}{r_I(t) + r_{I_0}(t)} = \frac{\zeta_I - \zeta_{I_0}}{2vt} + o(t^{-1}),$$

so (3.20), with $w(r_{I_0}(t)) \sim vt/\tau^2$, gives

$$\frac{\overline{F}(r_I(t))}{\overline{F}(r_{I_0}(t))} \longrightarrow \exp\left\{ -\frac{\zeta_I - \zeta_{I_0}}{2\tau^2} \right\}.$$

The remaining factors in (4.28) contribute $t^{-|I|}$, up to a common factor and a positive constant. Hence the largest scores and then the smallest cardinalities select exactly $\mathfrak{H}$. Consequently, for $I_0 \in \mathfrak{H}$ and $I \in \mathfrak{I}_*$

$$\lim_{u\to\infty} \frac{p_I(u)}{p_{I_0}(u)} = \frac{\widehat{A}_I}{\widehat{A}_{I_0}} \mathbf{1}_{\{I \in \mathfrak{H}\}}, \qquad \widehat{A}_I = \frac{\tau^{|I|} \exp(-\boldsymbol{\mu}_I^{\mathsf{T}} \Sigma_{II}^{-1} \boldsymbol{\mu}_I/(2\tau^2))}{(2\pi)^{|I|/2} \sqrt{\det \Sigma_{II}} \prod_{j \in I} \lambda_{I,j}}.$$

These are precisely the weights in (2.8) for covariance $\tau^2\Sigma$, proving the assertion about C_τ.

In part (ii), $h_I(t) \to \infty$, $h_I(t)/t \to 0$ and

$$\frac{\mathbf{\Delta}_I}{h_I(t)} = O_{\mathbb{P}}\Big((r_I(t)w(r_I(t)))^{-1/2}\Big) \longrightarrow \mathbf{0}_I \quad \text{conditionally on } H_I(u).$$

Thus (4.14) and the log-sum-exp inequalities give, for $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$,

$$\text{Law}\left(\left(\frac{\log(T_\ell/u)}{h_I(t)}\right)_{\ell\in\mathcal{C}_I} \mid H_I(u)\right) \xrightarrow[u\to\infty]{\text{w}} \text{Law}\left(\left(\max_{j\in J_{\ell,I}} Z_j^{(I)}\right)_{\ell\in\mathcal{C}_I}\right).$$

The limiting maxima have no atom at zero, and their joint positive orthant has positive probability. Hence sums and maxima have the same boundary factors. Summing the local contributions proves (3.22). □

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